

Changda Ma

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Research Interests: Spatial Analytics, AI for Urban and Architecture, Acoustic and Noise Environment, Spatial Sound, VR/AR, Music Audio Generation

EDUCATION

Georgia Institute of Technology, The US Master of Science in Music Technology and Master of Science in Architecture	08/2024-05/2027
Georgia Institute of Technology, The US Master of Architecture	08/2022-05/2024
Guizhou University, China Bachelor of Architecture	09/2017-07/2022

PUBLICATION

C Ma., S Wang., C Zhu., & A Smith. (2026). **Extending Xenakis: From Architectural Geometry to Sonification of the Philips Pavilion** In Proceedings of the International Computer Music Conference (ICMC 2026), Hamburg, Germany. (Accepted)

S Wang*, C Ma*, C Zhu*, G Mobarak*, & H Von Coler. (2026). **Immersive Open Studio: Shared Space, Shared Sound.** " In Proceedings of the International Conference on New Interfaces for Musical Expression (NIME 2026), London, UK, 2026. *Co-first authors. (Accepted)

D Hu., C Ma. & T Rakha. (2025). **An Integrated Framework for Rooftop Solar Viability Assessment: Enhancing Building Reuse and Urban Sustainability through UAV-Derived 3D Modeling and Performance Simulation** " Energy and Buildings, <https://doi.org/10.1016/j.enbuild.2025.116847> (Published)

C Ma., Y Shi., C Wang., & P Kastner. (2025). **Establishing a Fire Risk Map Based on Planned Spatial Layouts and Environment in Rural Planning: A Case Study of a Stereotype Village in China** In Proceedings of 27th International Conference on Human-Computer Interaction (HCI International 2025). (Published)

C Wang., Y Shi., C Ma., H Xu., & P Kastner. (2025). **Reimagining Chinese Garden Design: An Interactive Approach Using Stable Diffusion.** In Proceedings of The Association for Computer-Aided Architectural Design Research in Asia (CAADRIA) 2025. (Published)

Y Shi., C Ma., C Wang., T Wu., & X Jiang. (2024). **Harmonizing Emotions: An AI-Driven Sound Therapy System Design for Enhancing Mental Health of Older Adults.** In H. Degen & S. Ntoa (Eds.), Artificial Intelligence in HCI (pp. 439–455). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-60615-1_30 (Published)

RESEARCH EXPERIENCE

4CUBES: Integrating Menu-Based Network and Multi-Headset Streaming for Collaborative XR Musical Instruments Atlanta, The US

Researcher | Advisor: Prof. Henrik Von Coler 08/2025-05/2026

- Developed a LAN-based multi-user VR menu system, enabling real-time collaborative music performance within a shared virtual environment.
- Designed Swarm Cube interaction modules to enhance user engagement and expand backend musical control layers.
- Assisted in developing a Patch Bay-style signal routing workflow to manage mappings between multi-user inputs and musical parameters

AI-R Lab Atlanta, The US

Robotics Research Assistant | Advisor: Prof. Patrick Danahy 01/2026-05/2026

- Developed an AR4 robotic arm simulation and control workflow using ROS 2 Humble, MoveIt, RViz, and Gazebo for motion planning, visualization, and validation.
- Built Isaac Sim and Isaac Lab workflows for AR4 manipulation tasks, including target following, RMPflow-based control, and pick-and-place simulation.
- Integrated Intel RealSense D455 depth camera for color-based object detection and vision-guided grasping, supporting dual-arm AR4 collaborative manipulation.
- Designed reinforcement learning workflows using PPO and Actor-Critic methods, including reward functions and policy networks for robotic grasping and placing tasks.
- Established a sim-to-real workflow by connecting Isaac Sim on Windows with ROS 2 Humble on Linux through ROS Domain ID, and documented reusable tutorials for future lab teaching and project handoff.

Neuroarchitecture: Urban Built Environments' Impact on Mental Health – A Systematic Literature Review Atlanta, The US

Researcher | Advisor: Prof. Patrick Kastner and Prof. Shalaila S. Haas 08/2024-05/2025

- Conducted a systematic literature review on the relationship between urban built environments and mental health, including keyword strategy, database search, eligibility screening, and data extraction.
- Processed filtered papers using OCR and NLP methods, including NLTK and Word2Vec, to support text mining and semantic analysis.
- Screened titles and abstracts, developed eligibility criteria, screened full-texts, and drafted data extraction
- Visualized research patterns through cosine similarity, correlation matrices, hierarchical clustering, cross-correlation matrices, and 2D relationship projection.

Data-Driven Fire Risk Mapping in Rural Planning: A Case Study of Stereotype Villages Atlanta, The US

Researcher | Advisor: Prof. Patrick Kastner 01/2024-05/2024

- Proposed a data-driven fire-risk assessment framework integrating village spatial layout, microclimate, topography, and emergency infrastructure.
- Built a Grasshopper-based simulation workflow using Urbano, Decoding Spaces, Butterfly, Eddy3D, and Ladybug to evaluate evacuation accessibility, visibility, wind environment, and terrain slope.
- Applied BehavePlus 5.0 to calculate wind adjustment factors, flame length, and surface fire spread rates, generating fire-risk maps for rural planning and management.

- Collected evidence from global reports and studies to assess the emotional well-being of elderly during the COVID-19 pandemic.
- Developed different types of songs and using “mirtoolbox” in MATLAB to analyze the brightness, flatness, centroid, and spread of high-mid and low-mid, pulse strength and rhythmic clarity as well as novelty. Through the theoretical models of arousal and valence, connecting the obtained music feature data to the theoretical models
- Participated in setting up an Arduino-based setup for precise HRV measurement and incorporated strategies to minimize error and noise.
- Designed the UI of the Sound Therapy robot and the guardian mobile app.

PROFESSIONAL EXPERIENCE

Shokz

AI Product & Wearable Systems Intern

Shenzhen, China
05/2026-07/2026

- Developed an AI-adaptive music interaction using physiological and motion data from wearable devices to guide workout music generation.
- Connected heterogeneous wearable hardware to a company-developed sports-agent backend through multiple transmission pathways: SDK-based Bluetooth fitness bands, direct Bluetooth sensing from Polar H10, Wi-Fi-based Apple Watch data transfer, and Fitbit API integration.
- Mapped real-time exercise signals, including heart rate, HRV, motion, and activity data, into backend for adaptive audio and wellness applications.

Graduate Research Assistant Of Georgia Tech

AI and Robotics in digital fabrication

Atlanta, The US
01/2026-05/2026

- Set up AI-integrated robotic workflows for material handling.
- Assemble hardware and integrate sensors/feedback systems.
- Develop ROS-based robot programs, and design and build robotic end effectors.

Graduate Research Assistant Of Georgia Tech

Sustainable Urban Systems Lab led by Prof. Patrick Kastner

Atlanta, The US
01/2025-05/2025

- Developed data extraction from the filtered papers in “Neuroarchitecture ”Systematic Literature Review.
- Collected all words from the filtered paper in “Neuroarchitecture”Systematic Literature Review using “OCR Scanning”method.
- Conducted Natural Language Processing(NLP) model such as Word2Vec and NLTK for preprocessing among all extracted words.
- Visualized Text Mining results by Cosine Similarity Matrix, Correlation Matrix, Cross correlation Matrix, and 2D relationship Projection.

Graduate Teaching Assistant Of Georgia Tech

M.Arch Required Course “Media Modelling II” Fall 2024 Teaching Assistant

Atlanta, The US
08/2024-12/2024

- Evaluated the assignment on “Mathematical Surfaces + 3D Attractor” and “Building System”.
- Developed tutorials for “Enscape for Rhino” and “How to install grasshopper toolkit”
- Managed daily attendance and solve students' problems on grasshopper.(44 students across two teaching sections)

Lab Research Assistant Of Georgia Tech

I2CE Lab Research Assistant led by Prof. Rocker

Atlanta, The US
05/2024-07/2024

- Conducted 3D modeling and created layouts and sections for the Georgia Tech Art Square Urban Design Study
- Calculated areas (sqft) for various types and floor layouts.

Graduate Teaching Assistant Of Georgia Tech

M.Arch Required Course “Integrated Building Systems II” Spring 2024 Teaching Assistant

Atlanta, The US
01/2024-05/2024

- Conducted tutorials and provided guidance on Revit and Rhino, focusing on collaborative workflows.
- Assisted in identifying structural and construction-related issues.
- Organized and maintained weekly sign-up sheets for group reviews, overseeing 23 students across 6 groups.
- Created instructional content on "How to Collaborate in Teams in Revit", enhancing students' technical skills.

AWARDS & ACADEMIC SERVICE

- GTCMT Grant Awardee, *SynOrbit – The Dual-VR- View Interactive Music Performance*, Georgia Tech, 2026
- Peer Reviewer, *Landscape and Urban Planning*, Elsevier, 2026

SOFTWARE SKILLS

Design & Modeling	Rhinoceros (master), Sketchup (master), AutoCAD (master), Revit (master),SolidWorks (medium) Grasshopper (master)
Visualization & Rendering	Adobe Suite(Photoshop(master),Illustrator(master),Indesign(master)),V-ray(master), Enscape (master)
Programming Language	C/C+ +(medium), Python(master), Java(medium), Javascript (master), R(medium)
XR & Game & Interaction	Unity(master), Blender(master), TouchDesigner (medium), Max/MSP/Jitter (medium), Processing(medium), P5.js(medium)
Robotics & Simulation	ROS 2(medium), MoveIt(medium), Gazebo(medium), Isaac Sim(Medium), Isaac Lab/Sim(medium), Arduino(medium)
Audio Production & Sound Design	Ableton Live (master), FL Studio (master), Reaper(master), IEM Plugin(master)
GIS & Spatial Analysis	ArcGIS Pro (master), Arcgis Online(medium), Arcgis Storymap(medium)

LANGUAGE&INTEREST

Language	Mandarin(Native), English(Fluent)
Interest	Guitar, Electric Piano, Drum, Singing, Basketball